

Introduction To Computer Networks

Few important terms:

Host / End Systems: Computers or devices connected to a network
e.g. Laptop, mobile, PC etc.

Packet Switches: It is an **intermediate device** in a network that takes incoming chunks of data on one of its links & forwards them onto other link e.g. router.

Transmission Rate: The rate ~~at~~ or maximum speed at which a packet switch can transmit packets (data) onto a link.

- Measured in **bits per second** (bps or Mbps)
- It is in per unit time & between 2 devices

Access Network: It is the physical infrastructure that connects an end system to a network / Internet.

e.g. router. ~~modem~~

What is a Network?

A network is simply a collection of interconnected devices ~~or systems~~, humans or objects.

A computer network is a collection of computers & other hardware devices connected so that users can share hardware, software and data & can electronically communicate with each other.

Internet is the largest computer network in the world.

Computer Network Topologies:

A **network topology** refers to the physical or logical layout of a network which can have various elements (nodes, links, devices).

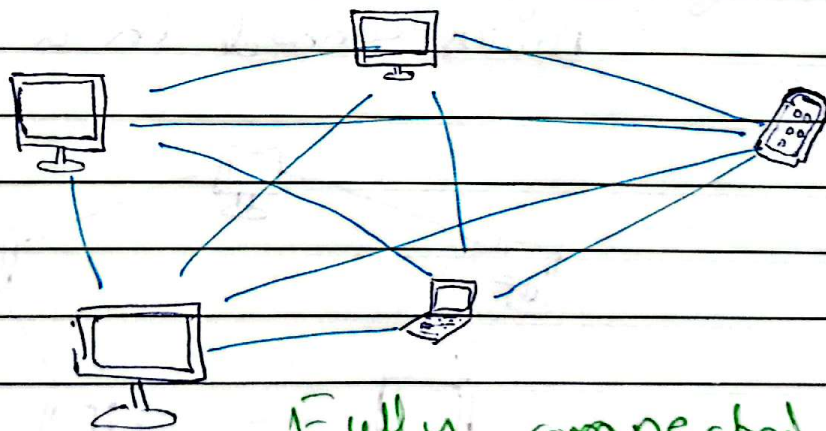
Topologies have multiple types.

Some Common ones:

- **Fully connected / Mesh:**

Devices connect to multiple (Partial mesh) or all other devices. (Full)

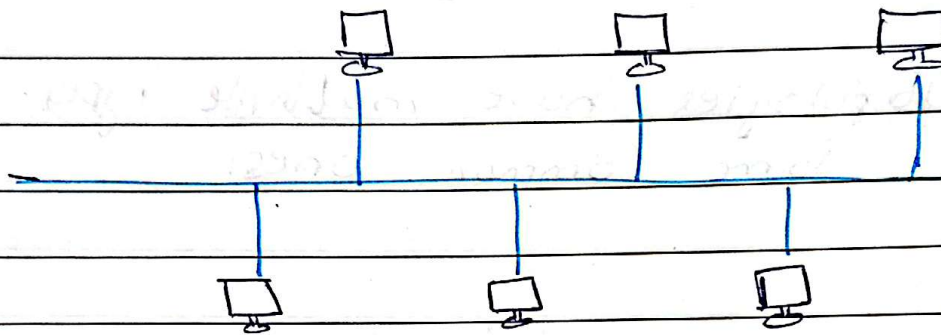
- It is extremely expensive & complex to cable & configure.



Fully connected.

• Bus:

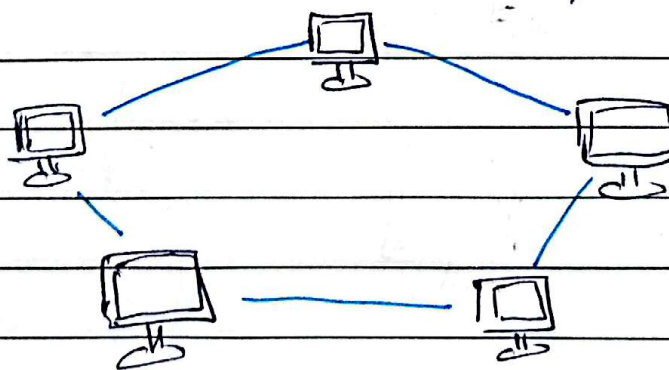
common "backbone" wire connecting every device.



- simple & cheap to set up
- Heavy collision risk as, if main cable breaks entire network goes down

~~Ring~~ • Ring:

Devices connect in a circular loop;



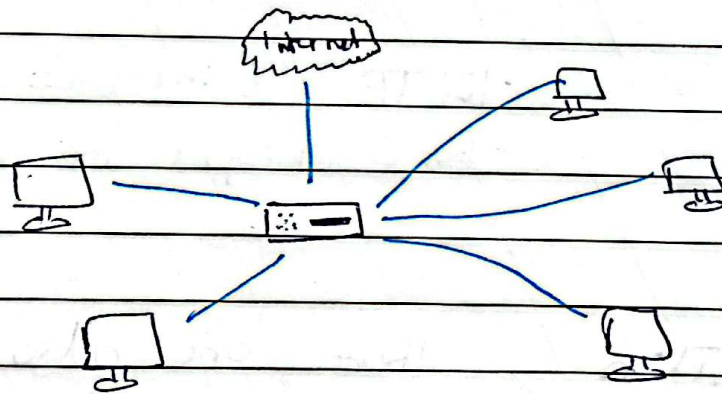
- No data collision; handles ¹ load data well without degradation.

- Failure in one cable ^{or device} can result in high delay times because longer path will have to be ~~take~~ taken.

Failure in 2 points will lead to a network breaking.

• Star :

Every device connects to a central hub or switch.



- Easy to manage or troubleshoot
- If central switch fails whole network goes down.

Internet:

Massive, global network of smaller computer networks linked together.

To understand it better we will divide it into 2 ways:

- 1) Nuts & bolts description (physical & technical hardware)
- 2) Services description (what it provides to end users)

1) Nuts & bolts description:

a) Millions of Computing devices (Hosts / End systems):

~~App~~

- Laptops / servers / smart TV's, sensor nodes
- They sit at edge & generate or consume the data transported across internet

Intermediate nodes:

These are **Hard ware devices** that take incoming blocks of data & forward them onto outgoing links towards final destination.

- e.g. routers, link-layer switches.

c) Communication Links:

Physical connection media that connects end systems & packet switches together

e.g. Fiber-optic cable, copper wire, radio spectrum.

• Transmission rate / bandwidth

d) Network of Network (ISP's).

Internet is a global collection of interconnected Internet Service Providers (ISP's)

e) Internet Standards

Standardized set of rules that ensure all Hardware & Software across Internet can communicate seamlessly regardless of device specs.

few common:

- RFC (Request for comments)
- IETF (Internet engineering Task Force)

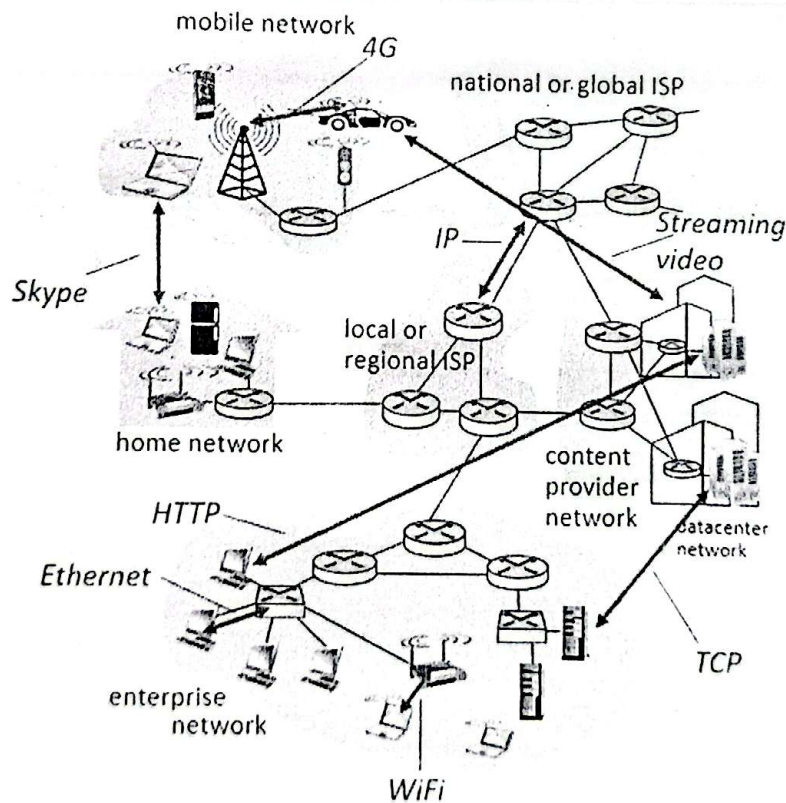
F) Protocols:

- Standardized set of rules that govern format, timing, sending & receiving messages throughout system.

To main protocols:

- TCP
- IP.

These protocols are standardized by the IETF & official documents called RFC.



2) services description:

a) Infrastructure

That provides services to apps e.g. web → streaming video, email, games etc.

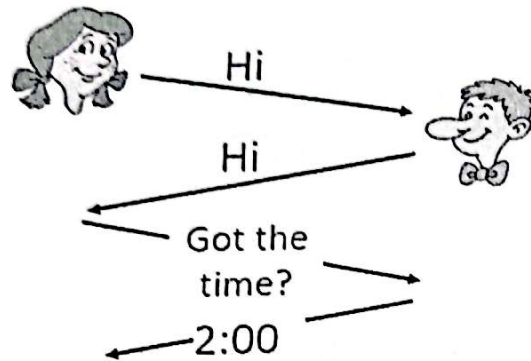
b) Programming interface for distributed apps:

- "books" allowing sending/receiving apps to connect to use Internet service
- provides service options, analogous to postal service.

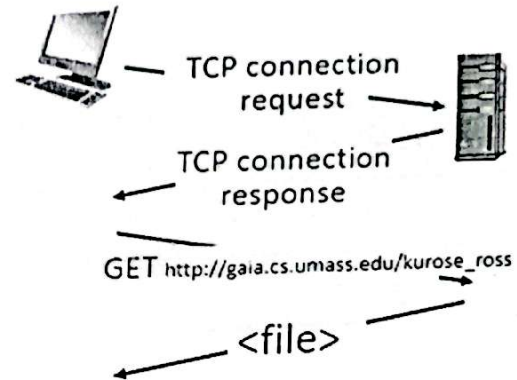
Protocols:

A protocol defines the format order & timing of messages sent & received among network entities, as well as the actions taken upon transmitting or receiving

A human protocol and a computer network protocol:



time



Closer look on Internet Structure:

Network edge:

- The ends of the network
- Hosts: clients & servers
- Servers: often in data centers.

- The end systems *horizontally*.

Access networks, physical media:

* will discuss more/in detail next.

- The links of the networks
- Wired wireless communication link

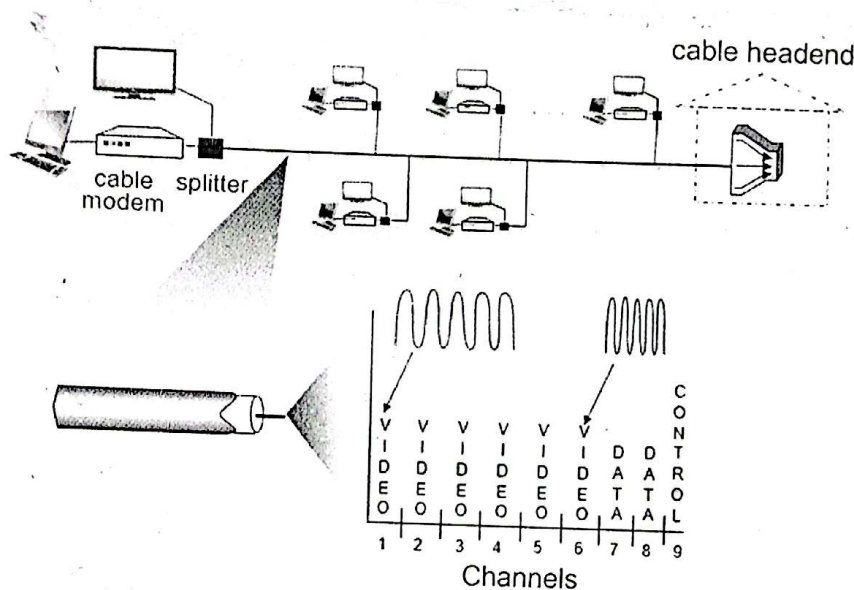
Network core

- This is the heart of network
- interconnected network
- Network of networks.

Access Networks (Retai Di:

Now access networks can be ~~of~~ of different types

Access networks : cable based access.



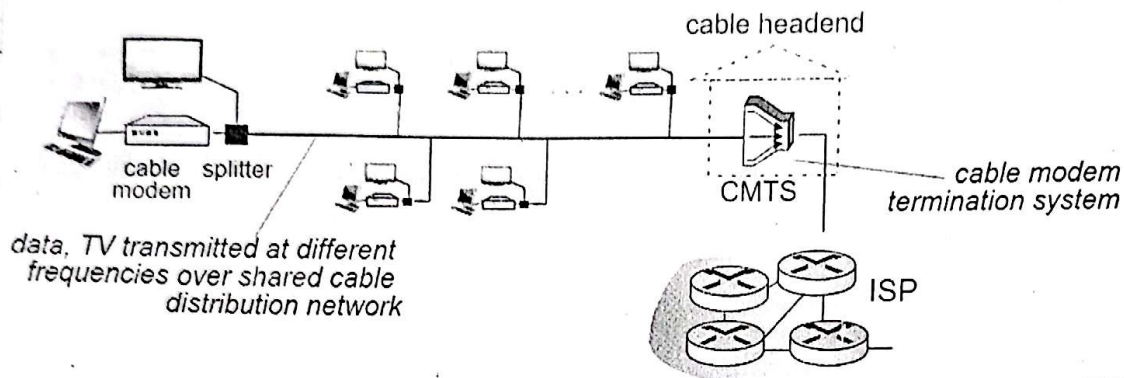
Cable networks use "Frequency Division Multiplexing"

Frequency Division Multiplexing (FDM):

The total frequency spectrum of cable is split into distinct, non-overlapping frequency bands.

Specific bands for specific stuff like TV channels.

day/date



• Hybrid fiber Coax (HFC):

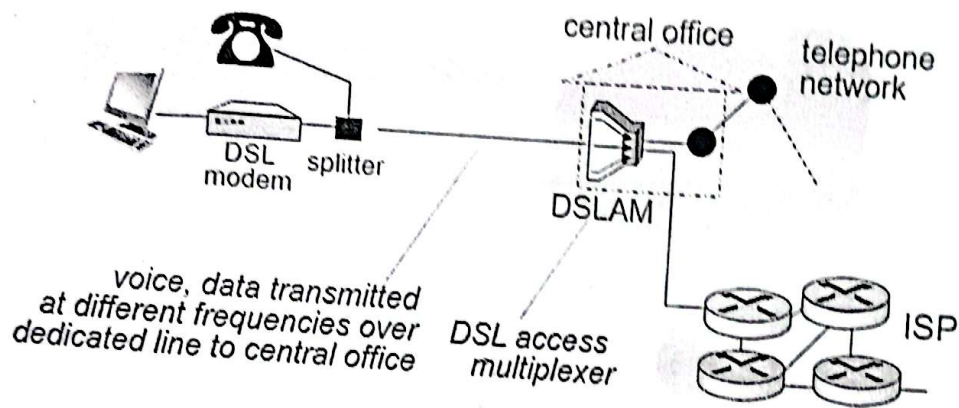
up to : 40 Mbps - 1.2 Gbps download
&
30-100 Mbps upload.

• Network of Cable.

• home share access network
to cable headend. (Top figure).

~~Network of Cable~~

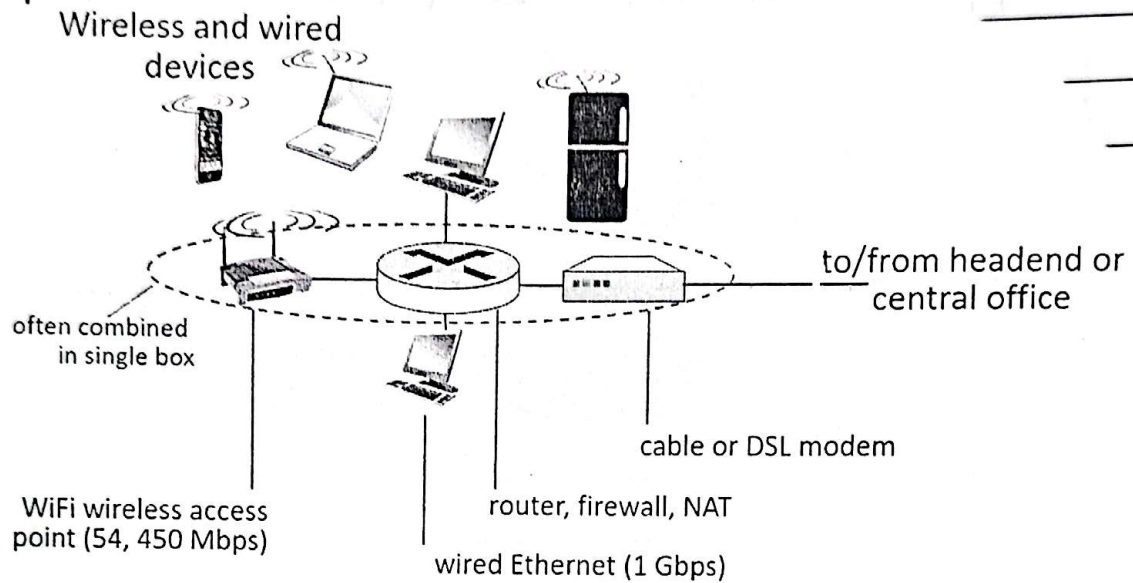
Access networks, digital subscriber line (DSL):



* DSLAM handles the telephone & data difference.

day/date

Access networks : Home Networks



Wireless Access network:

Shared wireless access network connects end system to router. via an access point.

- Wireless Local Area Network (WLAN):

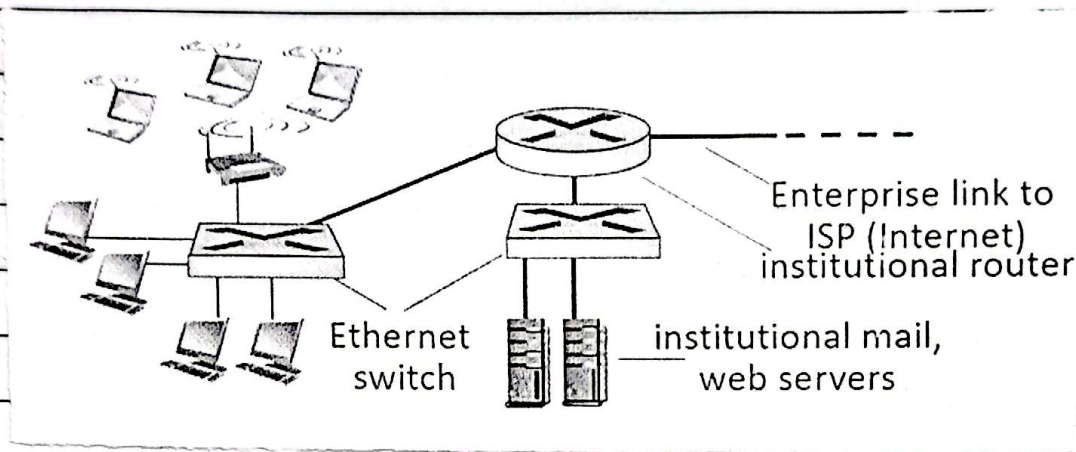
- ~100 ft ~~air~~ range
- ~~eg~~ ~~wifi~~ e.g. 80.211 b/g/n (w.i.f.i)
has 11, 54, 480^{mbps} bandwidth.

- Widearea cellular access network:

- provided by mobile, cellular network operator (10's km)
- 10's Mbps.
- ~~eg~~ e.g. 4G/5G networks.

Access network: Enterprise networks

- mix of wired, wireless link technologies, connecting a mix of switches & routers.



Access network: Data center networks

- high bandwidth links (10s to 100s Gbps).
connect hundreds to thousands of servers
together & to internet.

Links: Physical Media

Understand a few terms first.

- bit: propagates between transmitter / receiver pairs.
- physical link: what lies between transmitter & receiver
- guided media: signals propagate in solid media ^{e.g.} copper.
- unguided media: Signal propagates freely e.g. radio.

Guided Media

1) Twisted Pair copper wire:

- 2 insulated copper wires.

category 5: 100 Mbps, 1 Gbps ethernet
" 6: 10 Gbps ethernet

2) Coaxial cable:

- 2 concentric copper cables.
- bidirectional
- broad band:

• multiple frequencies on cable.

• 100's Mbps per second.

3) Fiber Optic Cable.

- glass fibre containing light pulses.
- high-speed operation (10's - 100's Gbps).
- low error rate.
- Immune to electromagnetic noise.
- repeaters spaced far apart.

Unguided media (wireless radio spectrum).

~~signals~~
wireless radio:

- signal carried in various bands in electromagnetic spectrum.
- no physical wire.
- broadcast "half duplex" (sender to receiver).

Radio link types:

- Wireless LAN (WiFi)
 - 10-100's Mbps; 10's of meters
- wide-area (e.g., 4G/5G cellular)
 - 100's Mbps (4G/5G) over ~10 Km
- Bluetooth: cable replacement
 - short distances, limited rates
- terrestrial microwave
 - point-to-point; 45 Mbps channels
- satellite
 - up to < 100 Mbps (Starlink) downlink
 - 270 msec end-end delay (geostationary)

Host: sends packets of data

The way data is sent by host is called host sending function:

- 1) take application message
- 2) break into smaller chunks known as packets of length " L " bits.
- 3) transmit packet across network at transmission rate R .

Packet transmission delay = time needed to transmit L -bit packet into a link

$$= \frac{L \text{ (bits)}}{R \text{ (bits/second)}}$$

* Remember transmission is between 2 devices (edges, routers, switches). but in an actual connection there can be many devices in between. So Total TD is

